GGG-A-576C
March 7, 1994
SUPERSEDING
GGG-A-576B
July 11, 1973

FEDERAL SPECIFICATION

ANVILS, BLACKSMITHS'

This specification is approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers anvils used by blacksmiths for forging metals into various shapes and sizes.

1.1.1 Federal specification coverage. Federal specifications do not cover all varieties of the commodity indicated by the title of the specification, or which are commercially available, but are intended to cover only those generally used by the Federal Government.

1.2 Classification.

1.2.1 Types. The anvils shall be of the following types as specified (see 6.1):

Type I - One horn
Type II - Two horns

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

GGG-T-563 - Tools, Blacksmiths'; Chisels, Flatters, Fullers, Set Hammers, Hardies, Bolt Heading Tools, Punches, and Swages
PPP-P-40 - Preservation and Packing of Hand Tools and Tool Accessories for Power Driven, Metal and Woodworking Machinery

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(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and Commercial Item Descriptions as outlined under General Information in the Index of Federal Specifications. Standards and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents. U.S. Government Printing Office, Washington, DC 20402.)

(Single copies of this specification, and other Federal specifications and Commercial Item Descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston, MA; New York, NY; Philadelphia, PA; Washington, DC; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Houston, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Seattle, WA.)

(Federal Government activities may obtain copies of Federal standardization documents and the Index of Federal Specifications, Standards and Commercial Item Descriptions from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage

(Copies of military specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM):

ASTM E 18 - Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

3. REQUIREMENTS

3.1 Illustrations. Except for dimensions, the illustrations herein are descriptive and not restrictive and are not intended to preclude the purchase of anvils otherwise conforming to this specification.

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3.2 Material. Anvils shall be of single piece cast steel, drop forged steel, or have a cast-iron body with steel face and horn, or horns, as applicable. The repair of defects in castings shall not be accomplished by peening, plugging, or impregnating. Defects in castings may be removed and repaired by welding at the discretion of the producer provided the deposited weld metal has a degree of hardness and other mechanical properties comparable to that of the parent metal. Castings shall be clean and free from injurious defects such as cracks, pits, and other defects which may affect serviceability of the anvils.

3.3 Design and construction. The anvils shall have a base, a face, and one or two horns as specified (see 6.1). The sides of the anvils shall be ground to an angle of 90 to the face for at least 1 inch below the top of the face. When anvils are furnished with a cast-iron body with steel face and horn(s), the face shall be of one piece steel having a carbon content of not less than 0.65 percent nor more than 0.90 percent. The face and horn table shall be securely welded to the body and shall remain true and not settle, break, or become loose from the body under normal service conditions. The working surface of the face shall be smoothly finished.

3.3.1 Hardie and pritchel holes. The anvils shall have a hardie (square) hole and a pritchel (round) hole which shall be smoothly finished. They shall be suitable for holding fullers, hardies, and swages as covered by GGG-T-563.

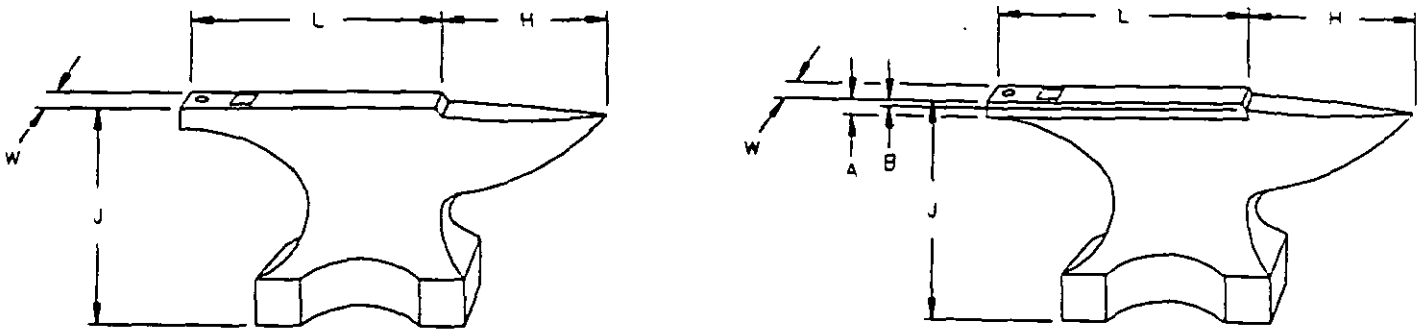
3.4 Hardness. The faces and horns of the anvils shall be heat treated to show a hardness of Brinell, Rockwell, or Shore Scleroscope as specified in table I when tested as specified in 4.4.1.

TABLE I. Hardness requirements

	Brinell 10-mm carbide ball 3,000 kg load		Rockwell		Shore scleroscope	
	Min.	Max.	Min.	Max.	Min.	Max
Face	500	588	54C	62C	72	85
Horn	203	344	93B	37C	31	50

3.5 Type I, one horn. The type I anvils shall be furnished with one horn, conform to the dimensions of table II for the weight specified (see 6.1), and be similar to figure 1.

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(All steel, single piece) (Cast-iron body, with steel face and horn)

Figure 1. Type I, one horn.

TABLE II. Types I and II anvils

*Weight		Heel A	Thickness of plate B	Square (hardie) hole	Round (pritchel) hole	Face Dimension L	Face Dimension W	Horn(s) (length) H	Height overall J
Min.	Max.	Min.	Min.	Min.	Min.	Min.	Min.	Min.	Min.
Pounds	Pounds	Inches	Inch	Inches	Inch	Inches	Inches	Inches	Inches
10	15	5/8	3/16	5/8	3/8	5	2	3-3/4	6
27	33	3/4	3/16	5/8	3/8	7-1/4	2-1/4	4-3/4	6
40	55	1	3/16	5/8	3/8	8-5/8	3	4-3/4	6-1/8
60	75	7/8	3/16	5/8	3/8	11	3-1/4	5-1/8	8
90	105	7/8	3/16	1	1/2	12-1/2	3-3/8	6-7/8	8
125	135	1-1/4	3/16	1	1/2	13	3-1/2	6-7/8	9
140	157	1-1/16	5/16	1	1/2	14-1/2	4	8	9
172	188	1-1/4	5/16	1	1/2	14-1/2	4-1/4	8	9
190	210	1-1/4	5/16	1-1/4	1/2	15	4-1/4	8	10
240	260	1-1/2	3/8	1-1/4	3/4	17	4-1/4	8	10
290	310	1-1/2	3/8	1-1/4	3/4	18	5	9	11

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3.6 Type II, two horns. The type II anvils shall be furnished with two horns, conform to the applicable dimensions of table II for the weight specified (see 6.1), and be similar to figure 2. If furnished with cast-iron body and steel face, thickness of plate shall be as specified in table II.

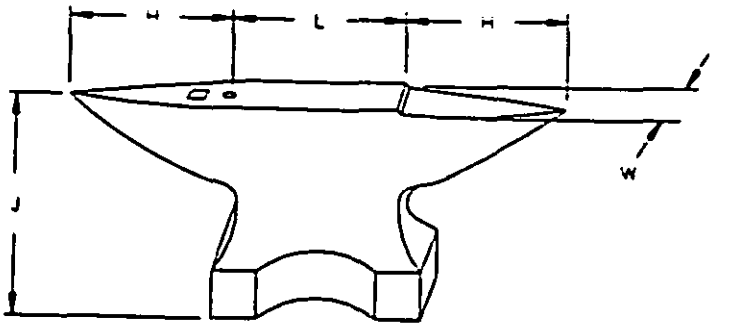


Figure 2. Type II, two horns.

3.7 Finished. Anvils shall be smooth and clean with all edges rounded or beveled. All surfaces not machined shall be painted or otherwise treated to resist corrosion in accordance with the manufacturer's standard practice.

3.8 Marking. Each item shall be marked with the manufacturer's name or identifying symbol and the state or country of manufacture. The marking shall be engraved, etched, or stamped in such a manner that it will be permanent to the extent that it will remain clear and legible throughout the life of the item.

3.9 Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the tolerances specified and all other requirements of this document are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request should be made to the specification preparing activity for change to this document.

3.10 Workmanship. The anvils shall be clean and free from defects which may impair serviceability, durability, or appearance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the

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performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. In accordance with 4.1, the supplier is responsible for insuring that components and materials used were manufactured, tested, and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified or, if none, in accordance with this specification.

4.2 Sampling procedures. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.3 Examination.

4.3.1 Visual examination. The end items shall be examined for the defects listed in table III. The lot size shall be expressed in terms of anvils. The sample unit shall be one anvil. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 4.0.

4.3.2 Dimensional examination. The end items shall be examined for any nonconformance with dimensional and weight requirements. The lot size shall be expressed in terms of anvils. The sample unit shall be one anvil. The inspection level shall be S-3 and the AQL, expressed in terms of percent defective, shall be 2.5.

4.4 Testing. The end items shall be tested for the characteristics specified in 4.4.1. The lot size shall be expressed in units of anvils. The sample unit shall be one anvil. The inspection level shall be S-3 and the AQL, expressed in terms of percent defective, shall be 2.5.

4.4.1 Hardness test. The sample anvils shall be subjected to a hardness test in accordance with ASTM E 18 to determine compliance with 3.4 and table I.

4.5 Inspection of preparation for delivery requirements. An inspection shall be made to determine that the preservation, packaging, packing, and marking comply with the requirements of PPP-P-40 and to section 5 of this specification. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

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TABLE III. Classification of preparation for delivery defects

Examine	Defect
Marking (exterior and interior)	Omitted: incorrect: illegible: improper size. location, sequence, or method of application
Materials	Any component missing, damaged, or not as specified
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling Distortion of container
Contents	Number per container is more or less than required Net weight exceeds requirements

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking shall be in accordance with PPP-P-40. The level of preservation, packaging, and packing shall be A, B, or Commercial, as specified (see 6.1).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Type of item required (see 1.2).
c. Level of preservation, packaging, and packing required (see 5.1).
d. Weight of anvil required (see table II).

MILITARY INTERESTS:

DOD waives coordination

Preparing activity:

GSA-FSS

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